

Alexander Oh

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Education

National University of Singapore, Singapore, SG
Bachelor of Science in Computer Science

Aug 2023 – Dec 2027 (Expected)

Professional Experience

Tetrix (Fintech Startup), New York, NY

Jan 2026 – Present

Backend Software Engineer

- Engineered and deployed an end-to-end regression pipeline tracing financial data coordinates; parallelized feature extraction to boost training throughput 30x on 50K+ documents, achieving 95.37% accuracy and 100% precision.
- Investigated failure patterns in complex edge cases through root-cause analysis, identifying feature gaps to strategically enrich the training dataset; engineered a multi-pass bootstrapping mechanism to capture surrounding data points as high-signal features, boosting edge-case accuracy by 35%.
- Spearheaded development of a scalable data extensibility framework and custom field system across 6 microservices, supporting 15+ private market data models while reducing schema change overhead by ~80%.
- Optimized large-scale financial computations using NumPy vectorization and built a DAG-based calculation engine with topological dependency resolution, improving time-series processing performance by 5–10x on 100K+ row datasets.

TikTok, Singapore, SG

June 2025 – Nov 2025

Backend Software Engineer for User Growth (Paid Ads)

- Designed a robust state-machine framework to manage complex payment approval workflows, cleanly decomposing 100+ conditional execution branches to minimize system edge cases.
- Eliminated payment duplication errors in distributed workflows by implementing async SQL locking mechanisms, achieving 0 duplicate approval incidents across 100+ per hour peak transactions.
- Implemented a high-throughput batch upload system with integrated deduplication and validation algorithms, leveraging asynchronous cron jobs to process 1,000+ concurrent files without impacting user-facing performance.

Asian Institute of Digital Finance (Fintech Research Institute), Singapore, SG

Jan 2025 – May 2025

Undergraduate Research Intern

- Automated generation of 3,000+ lines of SQL migration scripts using Python, reducing estimated manual migration effort from ~10 weeks to under 5 days and enabling zero-downtime schema transitions.
- Leveraged OpenAI APIs, fuzzy matching, and pandas to reconcile legacy database variables during Bloomberg migration.
- Refactored the Python backend to integrate new databases, rewriting outdated queries while implementing comprehensive error handling and multilingual input support.

Oneberry Technologies (Security Enterprise), Singapore, SG

May 2024 – Aug 2024

Fullstack Software Engineer

- Built a real-time monitoring platform capable of concurrently processing and streaming telemetry from 10,000+ live IoT sensors with low-latency updates (60k+ per hour).
- Led backend development using Node.js, RabbitMQ, MQTT, and Socket.io to build scalable event-driven messaging and websocket communication infrastructure.
- Deployed and maintained cloud-hosted services on DigitalOcean using PM2 process orchestration to manage concurrent workers, improving system reliability and supporting live production-scale testing.

Projects

Personal Website

Dec 2025 – Present

https://github.com/alexandtheoh/alexander_oh_website

- Developed a browser-native context aware RAG pipeline using WebLLM and Gemma-2 2B, leveraging TypeScript-implemented cosine similarity and top-k retrieval over 10K+ pre-processed indexed chunks.
- Engineered a conversation compression layer to dynamically prune and summarize chat state, maximizing token efficiency within strict client-side hardware limits.

Whatsapp LLM

May 2025 – Present

https://github.com/alexandtheoh/whatsapp_llm

- Fine-tuned a Mistral 7B language model on 250K+ conversational messages using QLoRA to explore personalized dialogue generation and conversational style adaptation.
- Designed custom conversational training templates and preprocessing pipelines to preserve multi-turn dialogue structure and long-range conversational context.

Skills and Interests

- Languages:** Python, Go, TypeScript, Java, C, SQL, VBA
- Libraries & Frameworks:** NumPy, Pandas, scikit-learn, Node.js, FastAPI, React, Thrift/Kitex, Hertz
- Systems & Infrastructure:** Docker, RabbitMQ, MQTT, Microservices, Socket.io
- Interests:** Tennis & Track (Represented NUS Club), Basketball, Golf, Hiking, Pokemon Showdown